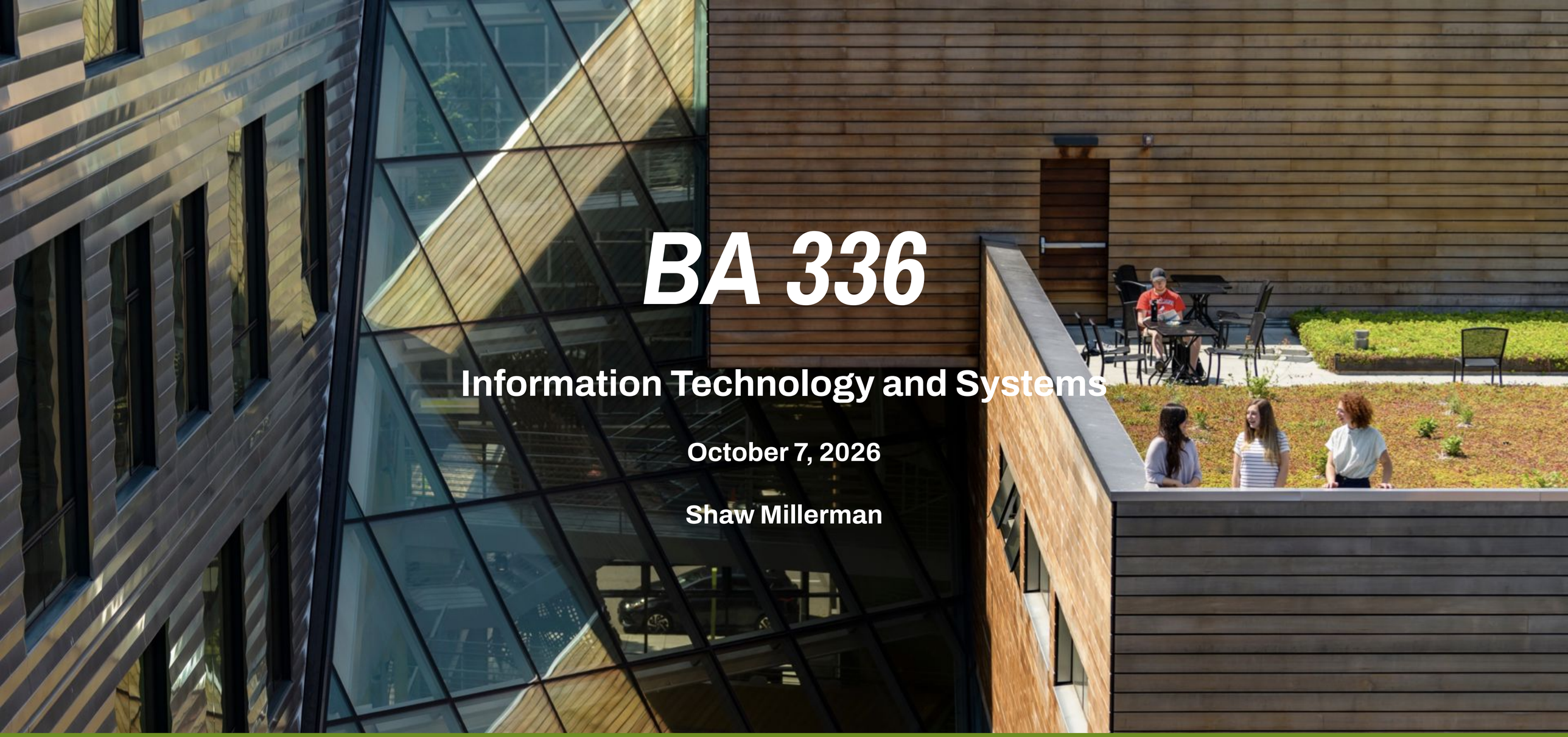


BA 336

Information Technology and Systems

October 7, 2026

Shaw Millerman

The School of Business
PORTLAND STATE UNIVERSITY

Agenda

- Welcome and Attendance
- Questions, Comments, Concerns
- Excel
  - Sheet Happens
  - Demo of Functions
    - Relative vs. Absolute
    - PMT
    - PMT Practice Sheet
- Moore’s Law and the Manager’s Dilemma
  -

Updates, Announcements, & What-Have-You's

- I've restricted the file type you can submit in Canvas. You might see errors.
  - xls
  - xlsx
- Put your name in the Excel file before you upload to Canvas
- Mistaken quiz announcement
- Excel textbook vs. the way you already know
- LinkedIn Extra Credit

Sheet Happens!

- I will show you an Excel sheet and you tell me what's off
- This is an exercise, not a personal attack.
  - We learn by finding and fixing small quirks that happen in Excel
  - If you made these same mistakes…it's okay
- Most of these are not "mistakes" but subtle tweaks:
  - Work looks more professional
  - Saves time
  - Prevent downstream errors
  - Stand out in the workplace
  - Our goal: to get better, together.



Sheet Happens!

Status Report Sales and Inventory by item						
Season	Item Number	Price		Item	Sales in Units	Inventory in Units
Fall 2018	A1500	$	89.99	Boots	130	1300
Fall 2018	A1501	$	79.99	Sweaters	350	2800
Fall 2018	A1502	$	49.99	Sweatshirts	1,800	10800
Fall 2018	A1503	$	29.99	Hats	1,000	7000
Fall 2018	A1504	$	129.99	Jackets	600	6000
Fall 2018	A1505	$	19.99	Scarfs	490	2940
Spring 2019	A1506	$	29.99	Shorts	1,400	5600
Spring 2019	A1507	$	19.99	T-Shirts	2,250	11250
Spring 2019	A1508	$	39.99	Sandals	850	3400
Spring 2019	A1509	$	12.99	Tank Tops	1,200	3600
Spring 2019	A1510	$	39.99	Polo Shirts	750	6000
		Totals			10,820	60690

Sheet Happens! (subtle edition)

Status Report *Sales and Inventory by Item*					
Season	**Item Number**	**Price**	**Item**	**Sales in Units**	**Inventory in Units**
Fall 2018	A1500	$ 89.99	Boots	$ 130	1300
Fall 2018	A1501	$ 79.99	Sweaters	$ 350	2800
Fall 2018	A1502	$ 49.99	Sweatshirts	$ 1,800	10800
Fall 2018	A1503	$ 29.99	Hats	$ 1,000	7000
Fall 2018	A1504	$ 129.99	Jackets	$ 600	6000
Fall 2018	A1505	$ 19.99	Scarfs	$ 490	2940
Spring 2019	A1506	$ 29.99	Shorts	$ 1,400	5600
Spring 2019	A1507	$ 19.99	T-Shirts	$ 2,250	11250
Spring 2019	A1508	$ 39.99	Sandals	$ 850	3400
Spring 2019	A1509	$ 12.99	Tank Tops	$ 1,200	3600
Spring 2019	A1510	$ 39.99	Polo Shirts	$ 750	6000
Total				$ 10,820	$ 60,690

Sheet Happens! (subtle edition)

Status Report
Sales and Inventory by Item

Season	Item Number	Price	Sales in Units	Sales Dollars	Inventory in Units
Fall 2018	A1500	89.99	130	11,699	1,300
Fall 2018	A1501	79.99	350	27,997	2,800
Fall 2018	A1502	49.99	1800	89,982	10,800
Fall 2018	A1503	29.99	1000	29,990	7,000
Fall 2018	A1504	129.99	600	77,994	6,000
Fall 2018	A1505	19.99	490	9,795	2,940
Spring 2019	A1506	29.99	1400	41,986	5,600
Spring 2019	A1507	19.99	2250	44,978	11,250
Spring 2019	A1508	39.99	850	33,992	3,400
Spring 2019	A1509	12.99	1200	15,588	3,600
Spring 2019	A1510	39.99	750	29,993	6,000
		Totals		413,992	60,690

Sheet Happens! (fixed)

Status Report *Sales and Inventory by Item*					
Season	**Item Number**	**Price**	**Item**	**Sales in Units**	**Inventory in Units**
Fall 2018	A1500	$ 89.99	Boots	130	1,300
Fall 2018	A1501	$ 79.99	Sweaters	350	2,800
Fall 2018	A1502	$ 49.99	Sweatshirts	1,800	10,800
Fall 2018	A1503	$ 29.99	Hats	1,000	7,000
Fall 2018	A1504	$ 129.99	Jackets	600	6,000
Fall 2018	A1505	$ 19.99	Scarfs	490	2,940
Spring 2019	A1506	$ 29.99	Shorts	1,400	5,600
Spring 2019	A1507	$ 19.99	T-Shirts	2,250	11,250
Spring 2019	A1508	$ 39.99	Sandals	850	3,400
Spring 2019	A1509	$ 12.99	Tank Tops	1,200	3,600
Spring 2019	A1510	$ 39.99	Polo Shirts	750	6,000
		Totals		10,820	60,690

Sheet Happens

Medical Office Budget					
Item	**Budget Cost**	**Quarter 1**	**Quarter 2**	**Quarter 3**	**Quarter 4**
Salaries	$ 180,000.00	$ 90,000.00	$ 30,000.00	$ 30,000.00	$ 30,000.00
Medical Supplies	$ 84,000.00	$ 42,000.00	$ 13,800.00	$ 9,000.00	$ 19,200.00
Medications	$ 126,000.00	$ 63,000.00	$ 20,700.00	$ 13,500.00	$ 28,800.00
Rent	$ 45,000.00	$ 22,500.00	$ 7,500.00	$ 7,500.00	$ 7,500.00
Office Supplies	$ 18,000.00	$ 9,000.00	$ 3,000.00	$ 3,000.00	$ 3,000.00
Phone & Utilities	$ 12,600.00	$ 6,300.00	$ 2,100.00	$ 2,100.00	$ 2,100.00

Sheet Happens! (subtle edition)

Medical office Budget					
Iteam	Budget Cost	Quarter 1	Quarter 2	Quarter 3	Quarter 4
Salaries	$ 120,000	$ 30,000	$ 30,000	$ 30,000	$ 30,000
Medical Supplies	$ 60,000	$ 18,000	$ 13,800	$ 9,000	$ 19,200
Medications	$ 90,000	$ 27,000	$ 20,700	$ 13,500	$ 28,800
Rent	$ 30,000	$ 7,500	$ 7,500	$ 7,500	$ 7,500
Office Supplies	$ 12,000	$ 3,000	$ 3,000	$ 3,000	$ 3,000
Phone & Utilities	$ 8,400	$ 2,100	$ 2,100	$ 2,100	$ 2,100

Sheet Happens: Pro-Tips

- Work looks more professional
- Saves time
- Prevent **downstream errors**
- Stand out in the workplace
- Our goal: to get better, together.



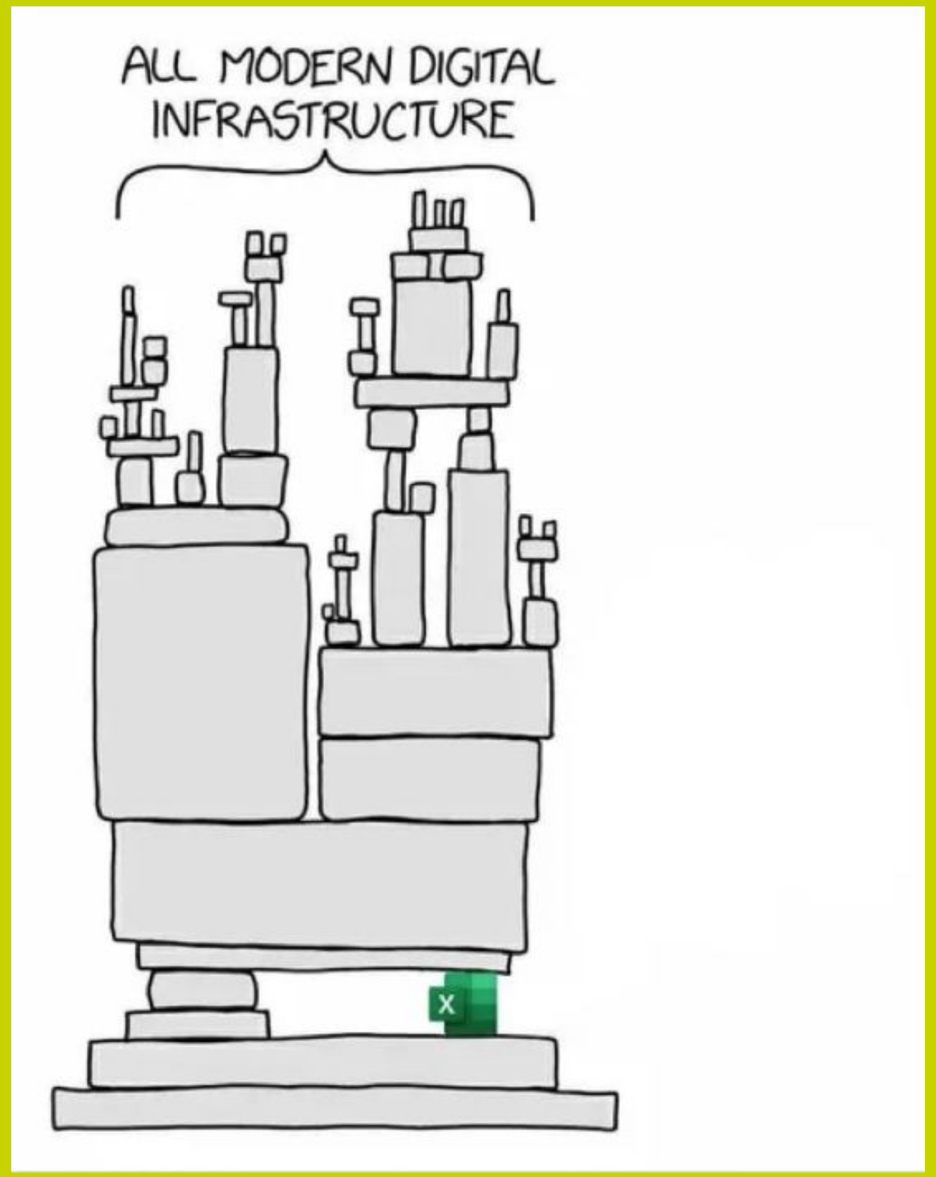
Excel Week 2

If Google Sheets is a bicycle, Excel is the race car

Modern Data Infrastructure

Still Runs on Excel

- Companies run on layers of tech
  - Some modern, some ancient
  - Software is/was rarely designed to speak together
- How does it happen?
  - Teams buy tools, vendors push integrations, and new tools and ideas
  - New leaders bring new platforms, new workflow, and new **data silos**.
- You ask question: “what customers are likely to cancel?”
  - What do you do?
  - **Manually download** data into _____.
- Global companies still running billion dollar divisions on nothing more than Excel
  - Download → Clean → Sort → Chart → Make decisions
- **Good news**: the tool we’re learning today isn’t going anywhere



Data Types (Using the Number Drop Down)

- Home Tab → Number Group
  - Number Format drop-down opens **quick formats**
  - “Launcher” arrow opens **full format options**
- Rules for Beginners
  - Don’t hand-type $
  - Don’t hand-type %
- Why it matters
  - Excel is smart but not perfect

Excel Data Types

Number	General	2354
	Number	2344.00
	Currency	$23,513.00
	Accounting	$ 4,234.00
	Date	3/4/2023
	Percentage	33.00%

Understanding Reference Types

Relative References

Default Behavior • Moves Relative to Copy Direction

```
= A1 + B1
```

When you copy this formula down or across, cell coordinates **automatically adjust** based on the offset.

- Copy down 1 row: `= A2 + B2`
- Copy down 2 rows: `= A3 + B3`
- Ideal for repeating calculations across rows.

Absolute References

Locked Coordinates • Uses $ as a Pin

```
= $A$1 + $B$1
```

The **$ symbol locks** the column or row that immediately follows it, preventing automatic changes.

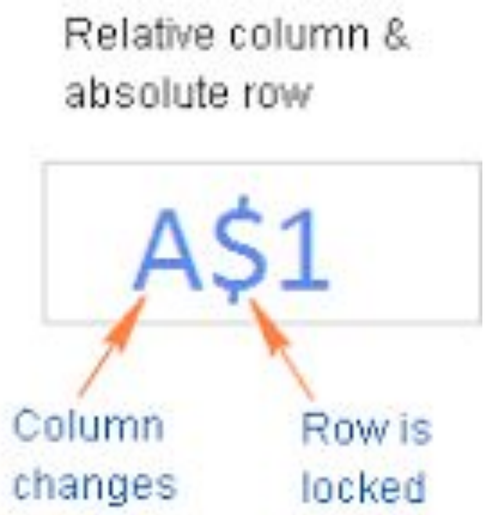
- Copy anywhere: `=$A$1+$B$1` (Unchanged)
- Lock Row Only: `=A$1` (Row 1 stays fixed)
- Crucial for referencing single lookup values or fixed rates.

💡 Key Takeaways & Shortcuts

- **The F4 Key:** Highlight a cell reference in the formula bar and press **F4** to cycle through absolute reference combinations (`$A$1` → `A$1` → `$A1` → `A1`).
- **Formula View:** Press **Ctrl + `** (tilde) to toggle show/hide all formulas on your spreadsheet to verify locked references.

Relative vs. Absolute References

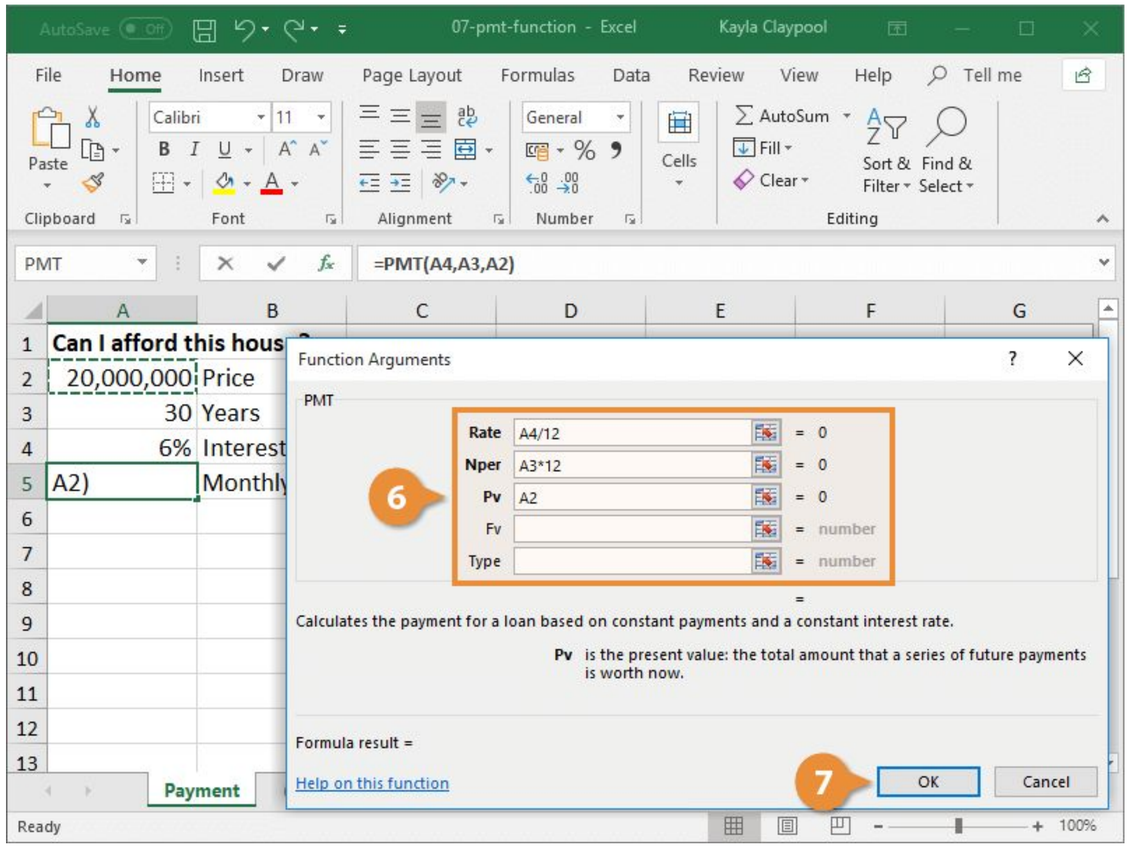




- Relative Reference in **copying formulas**
  - Adjusts to next row automatically
    - =A1+B1 → A2+B2
- Absolute Reference
  - Stays locked when you copy a formula
    - =$A$1+$B$1 → =$A$1+$B$1
- Think of $ as a pin
  - Locks the coordinate that follow
- Demo

PMT Function

- Calculates the payment for a loan
  - Interest rate
  - Length of the loan
  - Amount of the loan
- Every function:
  - starts with =
  - name function (PMT)
  - **arguments** (inputs needed to do the math)
- Arguments
  - Rate: interest rate per period
  - Nper: total number of payments
  - PV = present value: loan amount today
  - FV = future value: leave blank for this class
  - Type = when payments are due: leave blank



Calculating the Payment without Excel

To calculate a monthly payment (M) without a financial calculator or Excel, you have to solve for the present value of an ordinary annuity. The formula looks like this:

$$M = P\frac{r(1+r)^n}{(1+r)^n - 1}$$

Variable Key:

- M: Total monthly payment
- P: Principal loan amount
- r: Monthly interest rate (Annual rate divided by 12)
- n: Number of months (Years multiplied by 12)

Portland State
UNIVERSITY

Moore's Law & The Modern Manager

Strategy, Technology, and Analytical Thinking



BA 325: Information Literacy & Technical Competence for Business Professionals | Winter 2026

Instructor: Shaw Millerman, MBA | Portland State University

WEEK 3 COURSE CONCEPTS

Moore's Law is an Observation, Not a Law

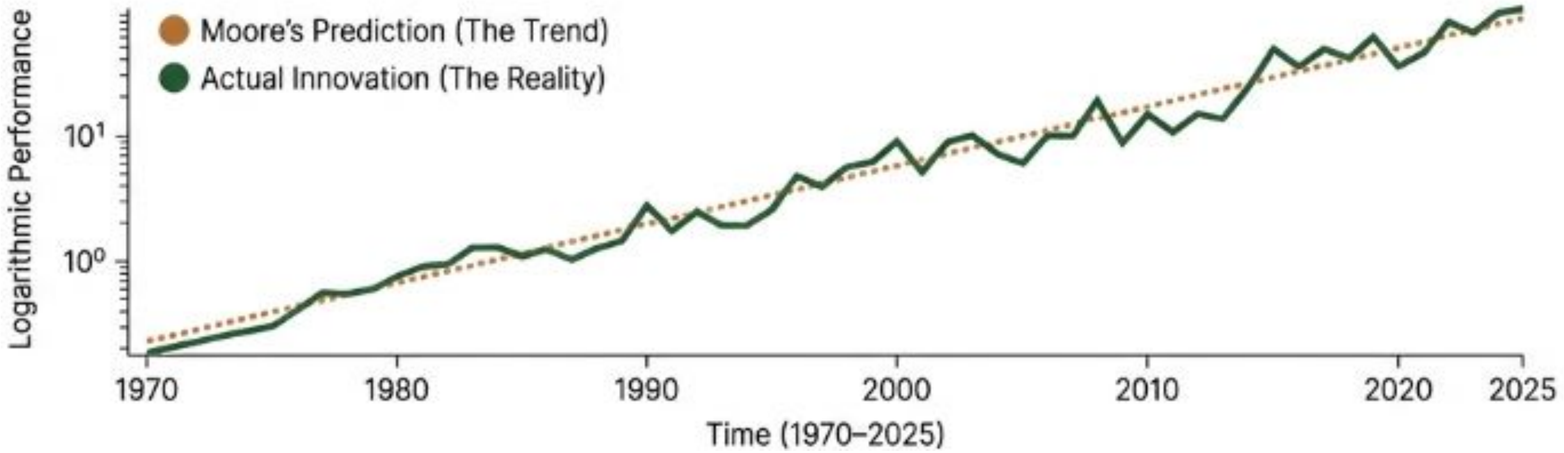
Gordon Moore (Intel Co-founder) observed in 1965 that transistor density doubled every two years.

Chip performance per dollar doubles every 18 months.

The Managerial Corollary: In 18 months, you can buy the same power for half the price, or twice the power for the same price.

Scope: Strictly applies to processors and chip-based storage (silicon).

The March of Innovation



Related "Laws"

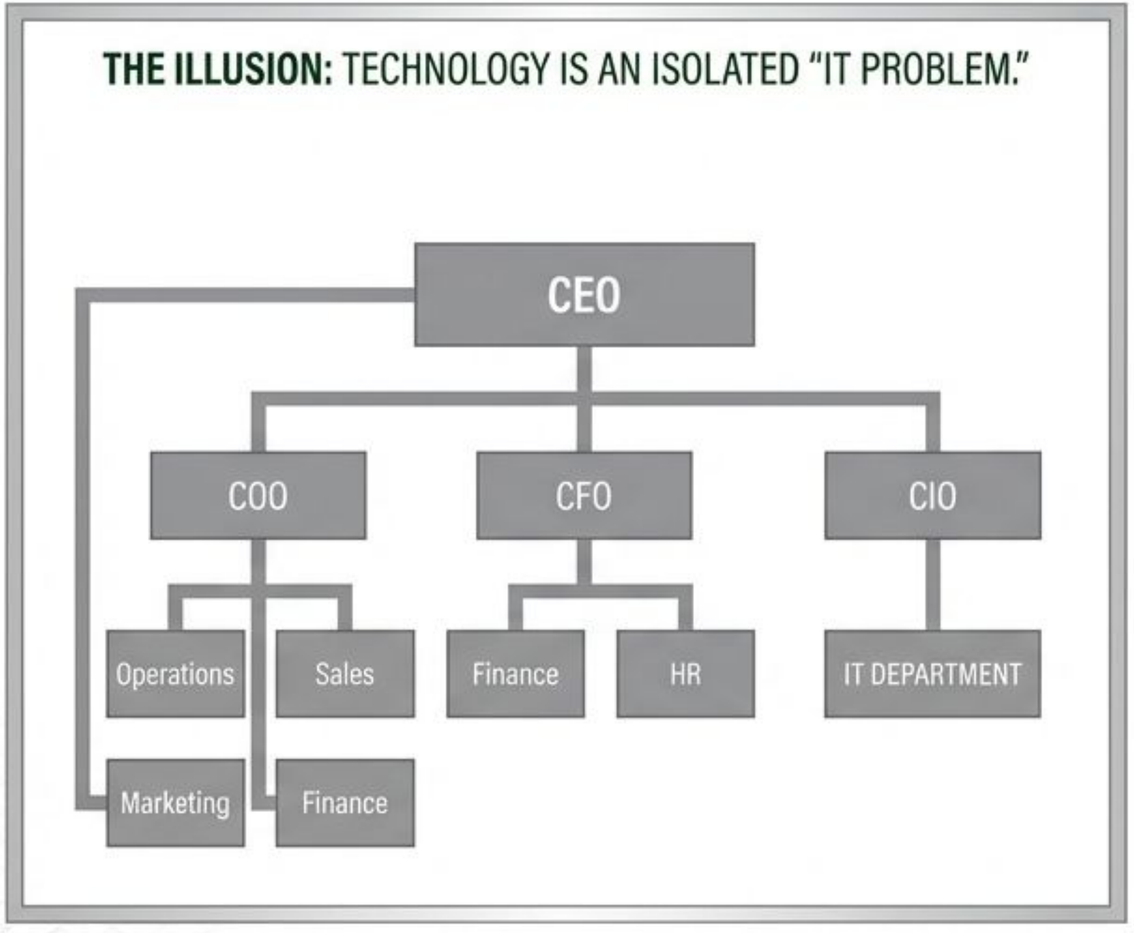
- **Data Storage:** Doubles every 12 months.
- **Optical Fiber:** Speed doubles every 9 months.
- **Kryder's Law:** Hard drive storage density.

A Manager's Job (In This Context)

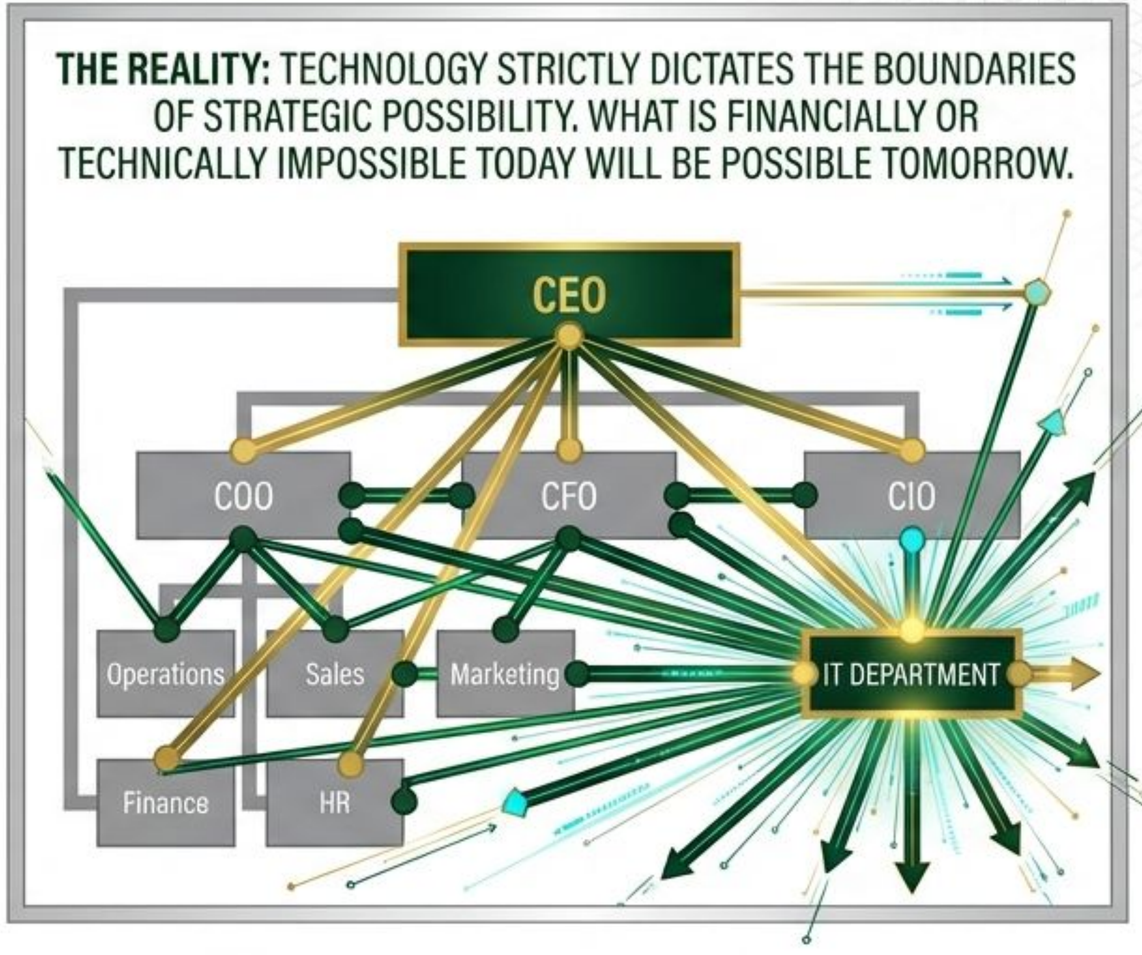


- Scan for new and emerging technologies (lower cost or improve product capability) e.g. Social Media Managers
- Run disciplined due diligence, problem, value, risk, cost to exit (**strategic analysis**)
- Decide **buy, build, partner, or wait**, based on evidence, not hype
- Implement change with people and process, training, support, incentives

THE MANAGER'S MANDATE



The Illusion: Technology is an isolated "IT problem."



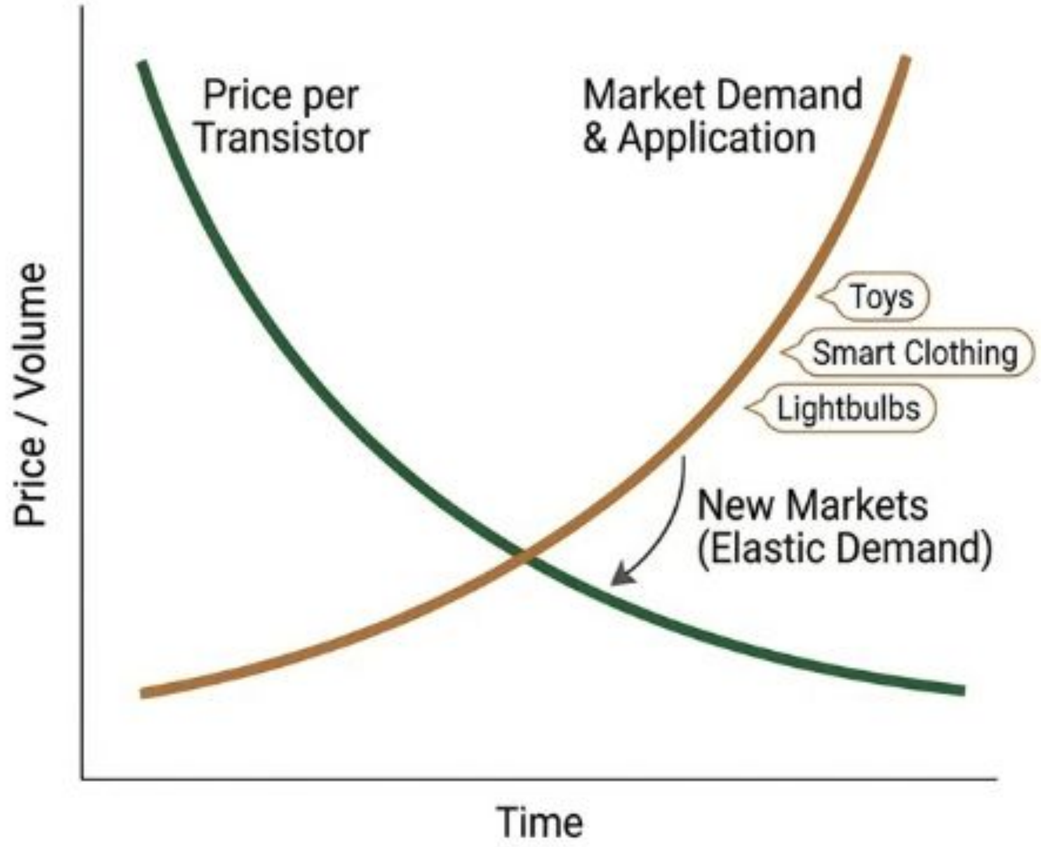
The Mandate: Leaders must continuously study technological trajectories to forecast price elasticity, recognize opportunity, and preempt industry disruption.

The Economics of "Faster & Cheaper"

Price Elasticity

Tech products are highly price elastic.

As chips get cheaper, consumers buy more, and entire new markets open up.

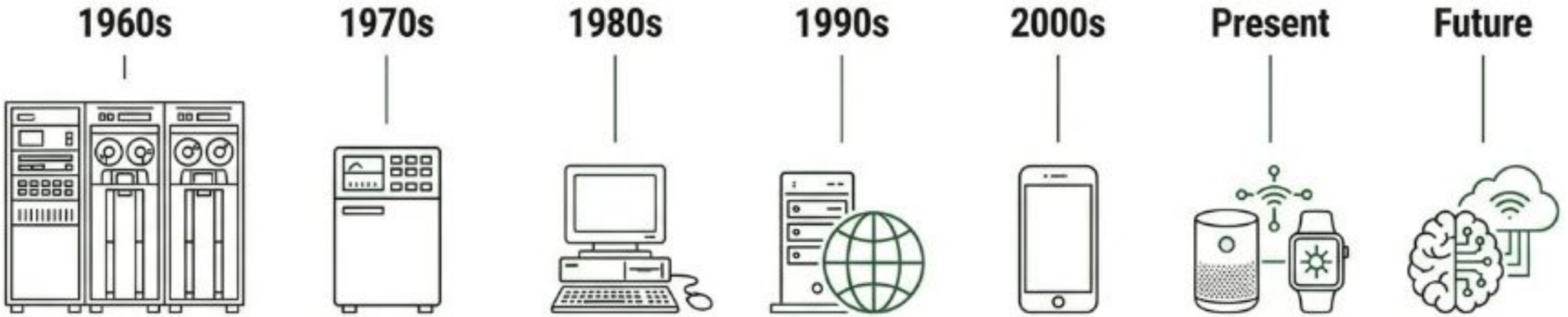


The Inventory Trap

Inventory depreciates ~1% per week (Dell Example).

Strategy: Carry minimal inventory to avoid holding devalued assets.

The Seven Waves of Computing



Key Insight: Successful firms surf these waves. Those that miss them become professional roadkill.

Editorial Strategy

Tech as a Global Equalizer



93% of the world has mobile broadband access—more than have access to clean water or electricity.

Case Study: M-Pesa (Kenya)

"Mobile phone as bank." 44% of Kenya's GDP flows through the service. 25 million users pay for groceries and tuition via text.

Case Study: Esoko (Ghana)

Solves Information Asymmetry. Farmers receive market prices via text. Result: Revenue increases of 10% to 40%.

The cell phone is the single most transformative technology for world economic development. — Jeffrey Sachs

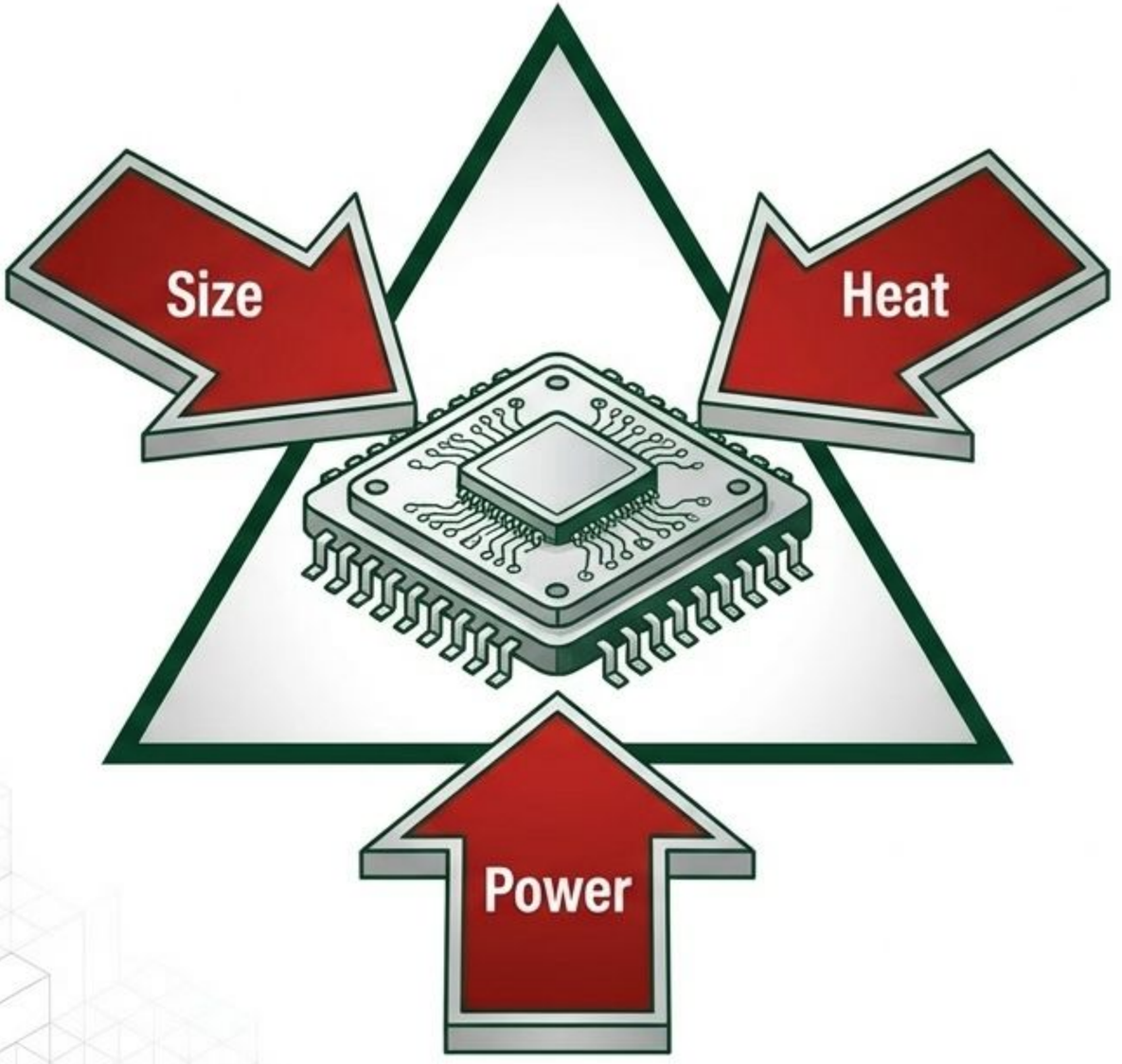
- Not one wave but two…
- 2007 Intel CEO Paul Otellini turns down Steve Jobs' request to build chips for the first iPhone due to disagreements over pricing and IP
- 2017, 2020, 2024 Intel assumes it's CPU can handle most computing tasks
  - Hello GPU and Nvidia

The Rise and Fall of Intel: How America's Largest Chipmaker Lost Its Way

Daniel Levi POSTED ON AUGUST 10, 2024

Share On Facebook Tweet It





The ‘Death’ of Moore’s Law?

Size: Transistor density is hitting the unyielding hand of physics. Current pathways are mere atoms wide.

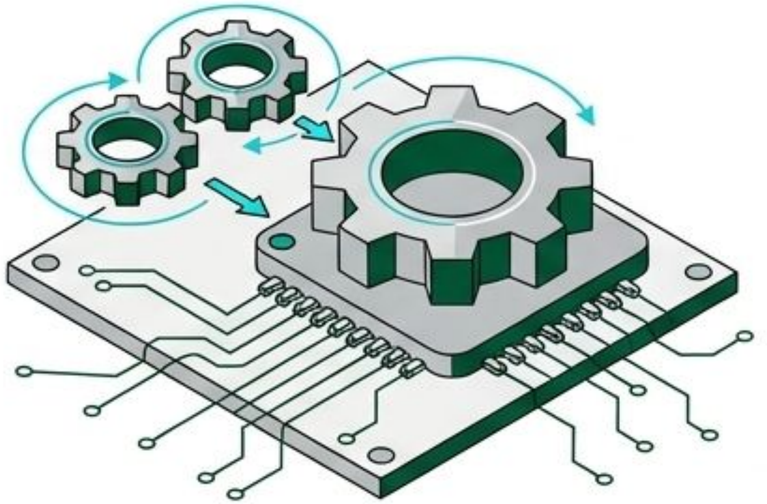
Heat: Tightly packed electrons generate massive thermal output, risking literal component meltdown.

Power: Cooling server farms now demands astronomical energy usage, pushing data centers to extreme environments like the Arctic.

The Shift: Squeezing chips into smaller spaces is no longer the sole path to faster computing. Innovation must pivot to architecture.

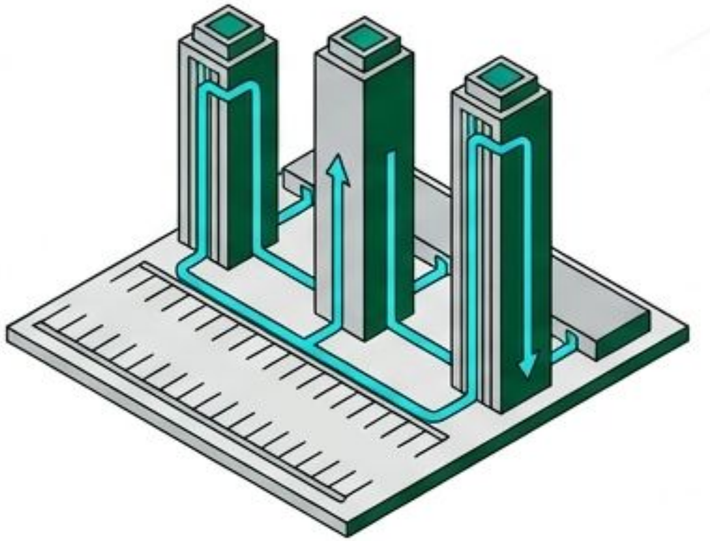
Architectural Shifts: Buying Time

Multicore Microprocessors



Multiple calculating cores on a single piece of silicon. Runs cooler and uses less overall power.

3D Transistors (RibbonFET)



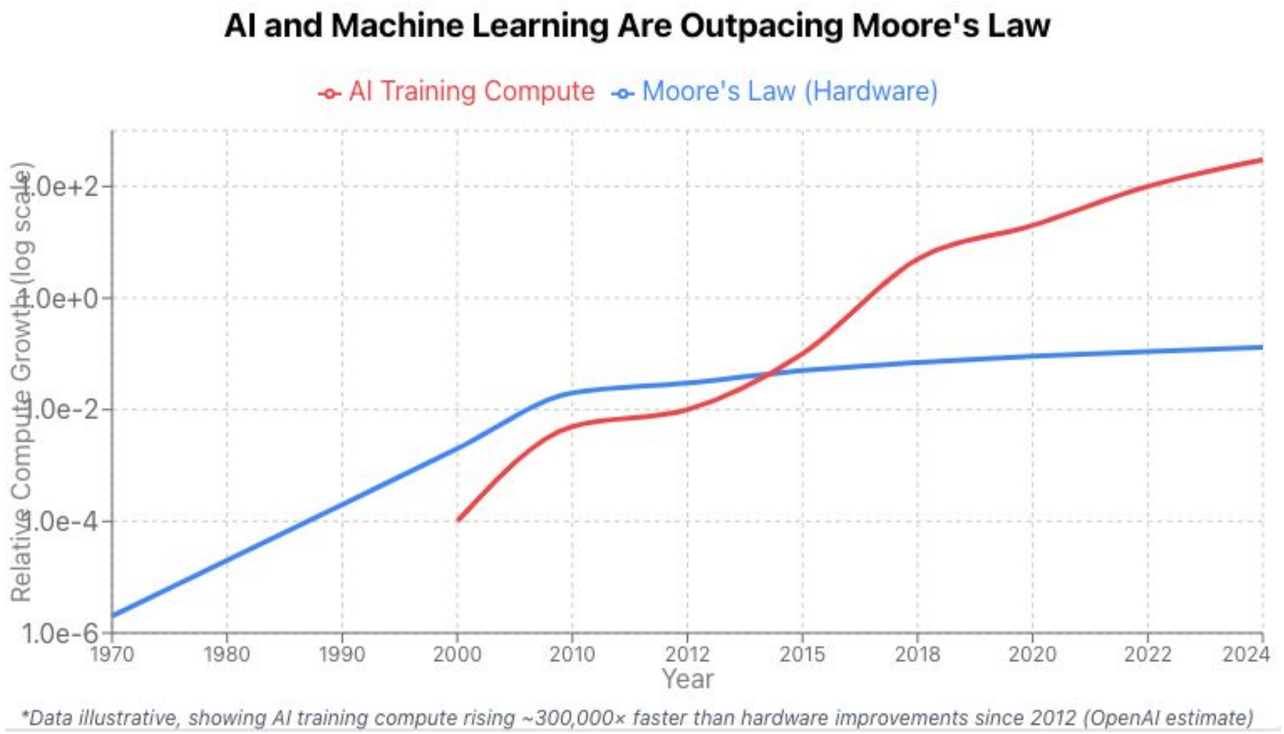
Shifting orientation vertically to reduce the distance electrons travel, saving space and massive amounts of energy.

The Switching Cost

Hardware is easy; but software codebases must be aggressively rewritten to divide and conquer tasks across multiple cores.

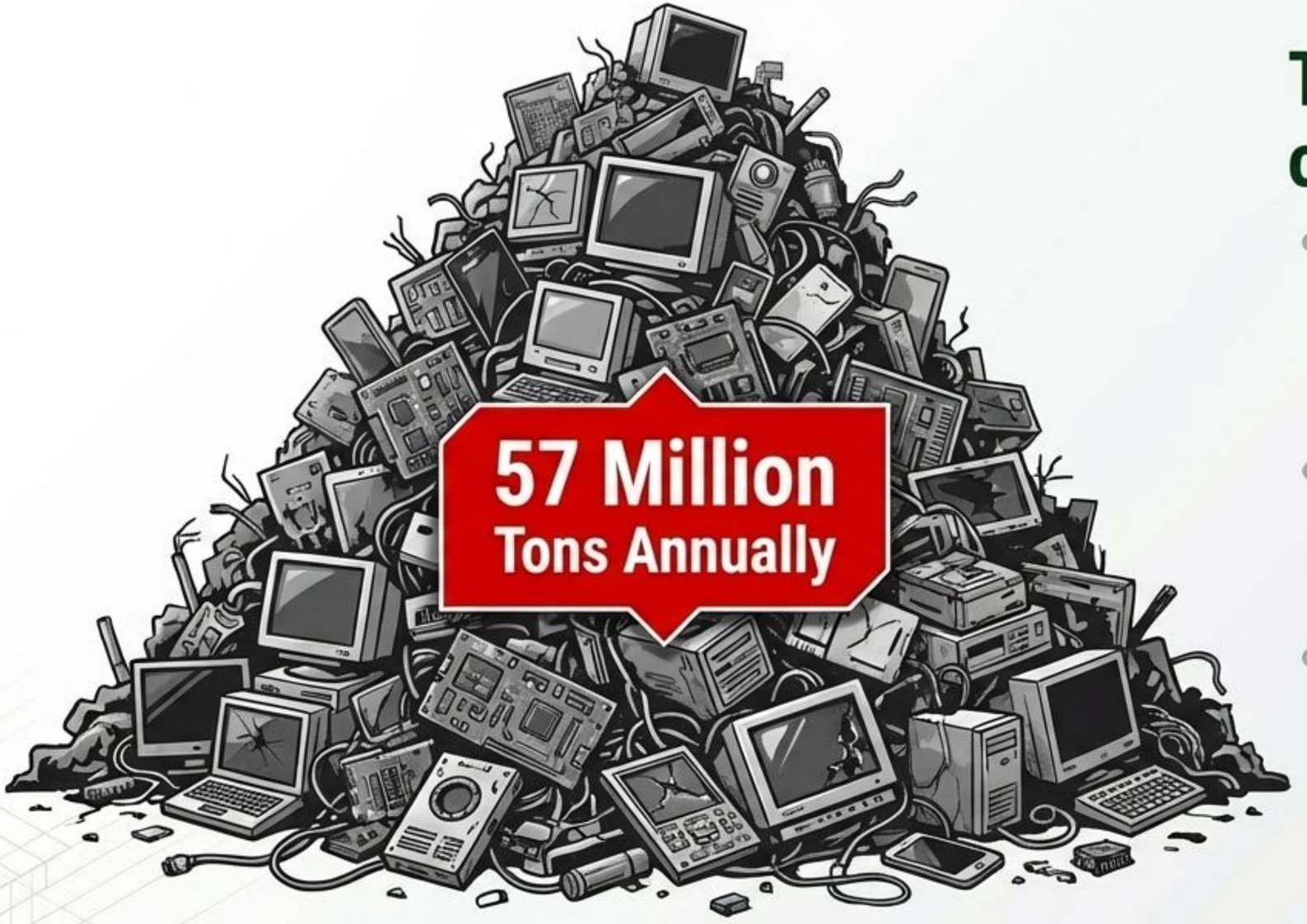
Moore’s Law Now

- Moore’s Law is slowing down but…
  - Squeezing more transistors onto a chip: not only path to faster/cheaper
- Chip design (where transistors go on chip)
- Cloud computing (e.g. Siri helps out)
- Superclusters
- Is Moore’s Law showing signs of its age?
  - Yes but the spirit is alive and well
- Rapid capacity growth requires constant adaptation (writing software, vendors to use,





- Between 2012 and 2022, AI brain power increased by 300,000x
- During same time, chips became 15x more powerful
- **Secret**: started linking thousands of chips together into massive "Superclusters" and invented clever new software tricks.
- The curve isn’t just steeper, it’s diverging exponentially from an exponential

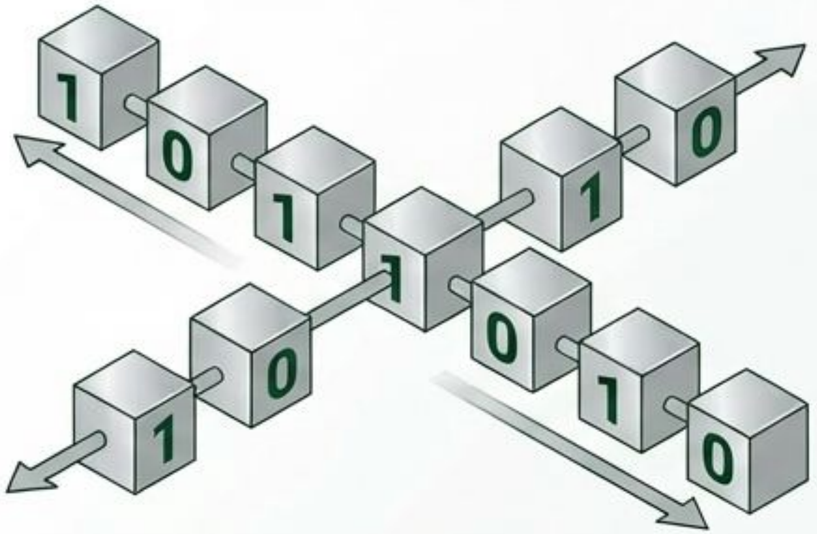


The Dark Side of Scale: E-Waste

- **The Hidden Cost:** Rapid technological obsolescence (Moore's Law) generates mountains of toxic e-waste.
- **The Resource Paradox:** A single ton of discarded iPhones contains more precious metals than 2,000 tons of newly mined ore.
- **Global Impact:** Exporting this waste to the Global South creates profound human health crises and environmental devastation.

Beyond Silicon: Quantum Computing

Classical Computing



The Leap: Leverages quantum physics using “qubits” that exist in superposition, shifting capacity from linear growth to exponential explosion.

Quantum Computing



Strategic Potential: Conventional computing requires an impossible **10^94 bytes** to model a serotonin molecule. A quantum computer requires precisely **424 qubits**.

Future Disruption: Hyper-detailed pharma simulations, unbreakable encryption, and complex logistical rerouting.



Case Study Intro: Disney's Billion-Dollar Pivot

- **The Crisis:** "The most magical place on earth" was losing its magic. Long lines, pricey tickets, and plummeting return-visitor intent.
- **The Hypothesis:** Friction destroys the brand experience.
- **The Investment:** A $1 Billion analytics ecosystem (MyMagic+) to aggressively remove operational friction using pervasive computing.



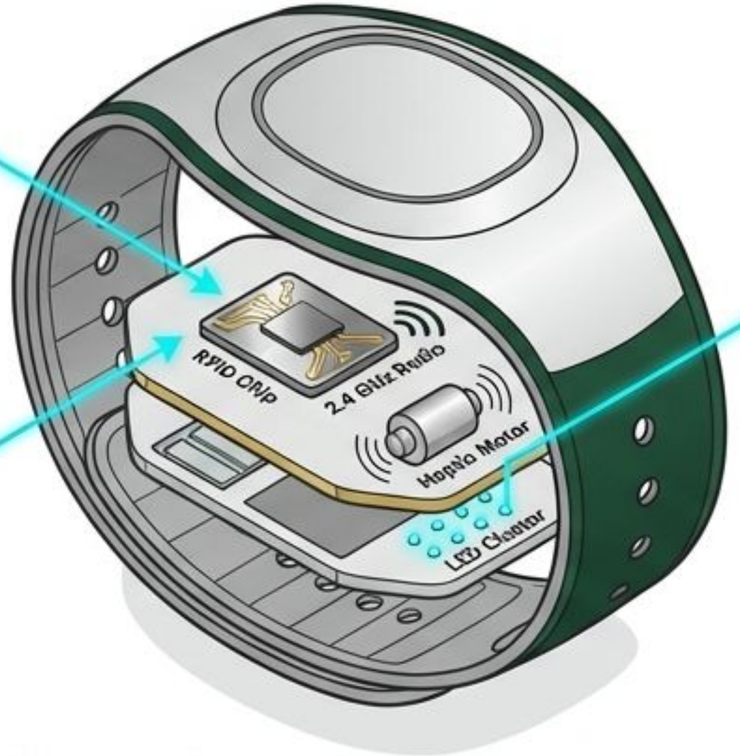
Disney MagicBand: WHAT (The Ecosystem)



Transactional:
Purchases, hotel room entry.



Logistical:
Lightning Lane reservations, ride check-ins.



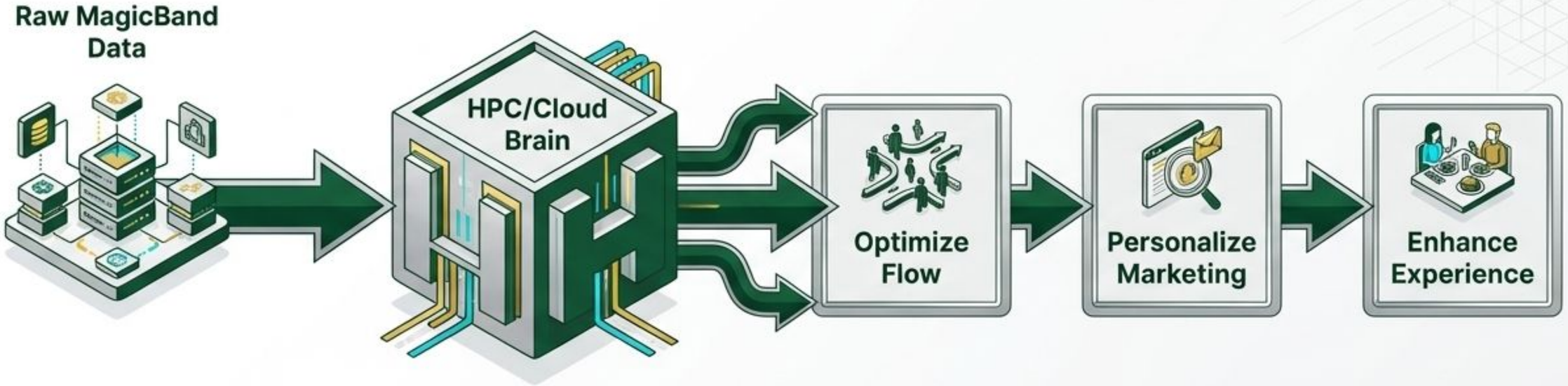
The Hardware:
A low-cost wearable (driven by Moore's Law) utilizing RFID, 2.4 GHz radios, haptics, and LEDs.



Movement:
Real-time park crowd flow tracking.

The Data Capture:
Generates roughly 5,000 data points per guest per 90 minutes.

Disney MagicBand: WHY



Optimize Flow: Sensors detect **bottlenecks**; **algorithms** dispatch character parades to actively **redirect crowd mass** in real-time.

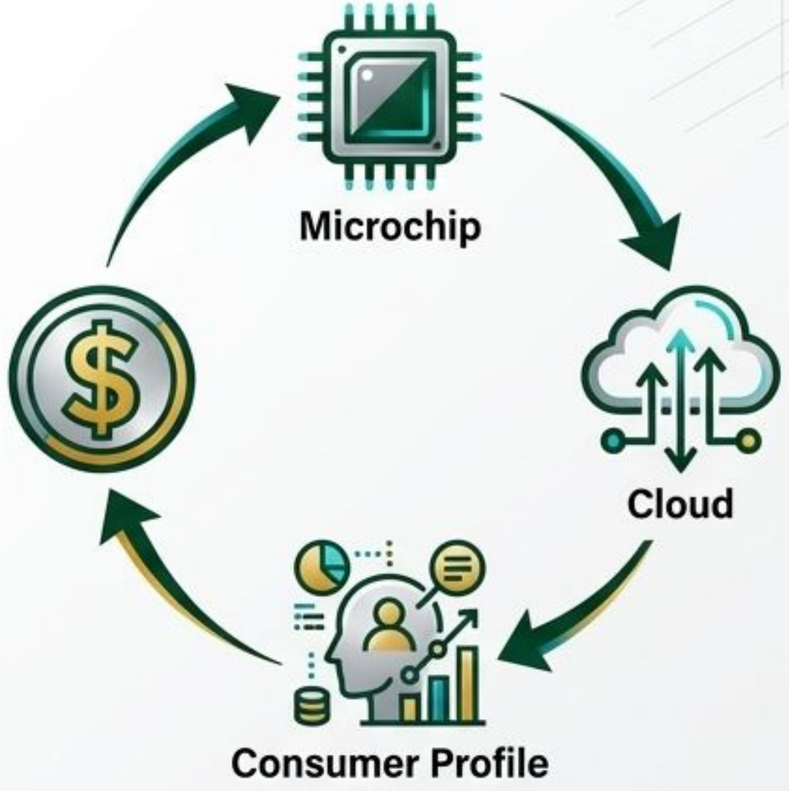
Personalize Marketing: Deep **behavioral insights** drive **targeted, high-conversion offers**.

Enhance Experience: Enabling "**invisible magic**" where food **intercepts guests perfectly** at their table.

The Reality: "It's more data than a human is going to look through. That's where computers come in." — Peter Carr, Disney Scientist

Disney MagicBand: RESULT (Competitive Advantage)





Operational Yield: Cut turnstile transaction times by **30%**, increasing park capacity by **3,000–5,000** guests daily.

Financial Yield: Frictionless tap-to-pay actively divorces guests from the psychology of spending, increasing overall revenue.

Strategic Yield (Porter): Deepens unparalleled brand differentiation, engineers massive switching costs for consumers, and widens the competitive moat against rivals like Universal Studios.

Synthesis: The Tech-Strategy Connection



Takeaways

- Moore’s Law is a strategic engine
  - Primary driver of price elasticity
  - Makes impossible possible
- Managers must navigate the seven waves of disruption
  - Firms that dominate in one wave, often fail to transition to the next. (Skate where the puck is going).
  - Which wave are we in now?
- Quantum is on the horizon but still lives in the world of theoretical
- Modern computing must be balanced with ethics, security of data, manage waste, etc.



A manager’s job is to convert falling compute costs, delivered through new software tools, into measurable business outcomes, while keeping people, data, and **exit options** under control.



Class Discussion: Disney & The Future of Data

Disney's MagicBand constructs a seamless, frictionless experience by extracting tens of thousands of data points regarding your physical movement, financial purchases, and behavioral habits.

The Prompt:

As a manager balancing Strategic Advantage with Corporate Responsibility, where is the line between "personalizing an experience" and "invading customer privacy"? How does a modern firm ensure they do not cross it?


Buh Bye